FIFTH ANNUAL UW–MADISON SUSTAINABILITY SYMPOSIUM

# When Does Better AI Actually Save More Energy? Decision-Aware Adaptive Fidelity for Building Control

A controlled precision-allocation study with implications for energy-aware building control

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## 01 Motivation: precision and building energy

Building controllers choose HVAC and equipment actions using sensor data, learned predictions, and operating constraints. More computation can refine a prediction, but whether it improves the final action depends on the decision rule. Selective precision may avoid costly inference where extra detail changes nothing.

**Research question: When does extra precision change a decision enough to justify its additional energy cost? We begin with controlled visual decisions before evaluating building control.**

## 02 Controlled complete-lattice design

A four-stage CNN trained on generated camera scenes is evaluated at all $2^4 = 16$ simulated activation-precision masks using the same frozen checkpoint and paired images.

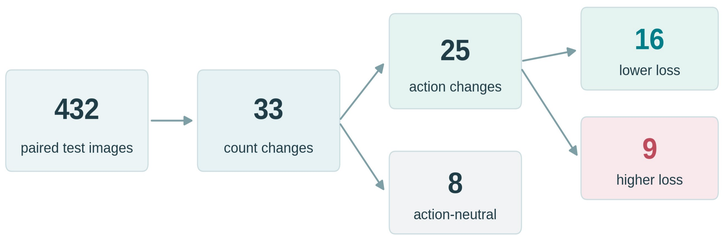
STAGE 1 FP / QDQ → STAGE 2 FP / QDQ → STAGE 3 FP / QDQ → STAGE 4 FP / QDQ

| 18 | 6 | 6 |
|---|---|---|
| train scene groups | validation groups | test scene groups |

Each group contains 72 frames; the held-out test set comprises 432 images at 32 × 24 resolution. Three-bit QDQ is simulated through floating-point operations, not native INT8.

## 03 Prediction changes versus action changes

The action is determined from the predicted count by the fixed mapping: counts 0, 1, 2, 3, 4 → actions 0, 1, 2, 2, 3. Thus some count flips leave the action unchanged.




All-low versus all-high differs on 33 predicted counts; 25 alter actions. Among those 25, moving to high precision improves toy loss on 16 images and worsens it on 9.

## 04 Evaluation unit and inferential limits

- All 16 masks were scored on the same images. Configurations are paired, not independent samples.
- The six held-out scene groups are the primary resampling unit; only one CNN training seed was evaluated.
- The three-bit QDQ interventions are simulated in floating point; GPU energy and native speed are not measured.

| 6 | → | 432 | → | 6,912 |
|---|---|---|---|---|
| held-out groups | | paired test images | | image-mask scores |

Derived evaluation count = 432 images × 16 precision masks; not 6,912 independent samples.

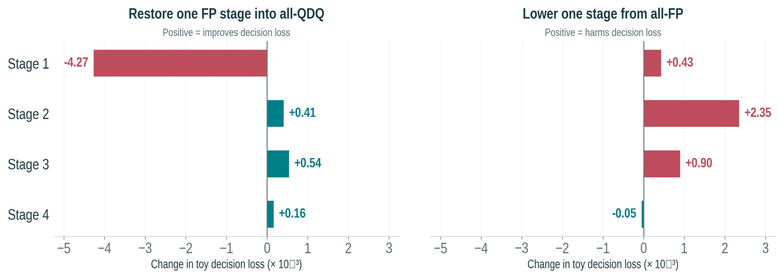
## 05 Relevance to adaptive building control

In a control loop, minor changes in occupancy or equipment-state estimates may leave an HVAC setting unchanged. Near a decision boundary, the same size prediction change can select a different operating action.

A viable controller must evaluate the quality of that action alongside inference joules, rerun frequency, comfort constraints and resulting facility energy use.

## 06 Restoration and lowering are asymmetric

Restore one stage in an otherwise low-QDQ graph, then separately lower that stage in an otherwise high-precision graph. These are distinct interventions: their effects can differ in sign and magnitude.




**Stage 1 restoration increased toy loss by 0.00427; six-group bootstrap interval for extra loss: 0.0021–0.0063.**

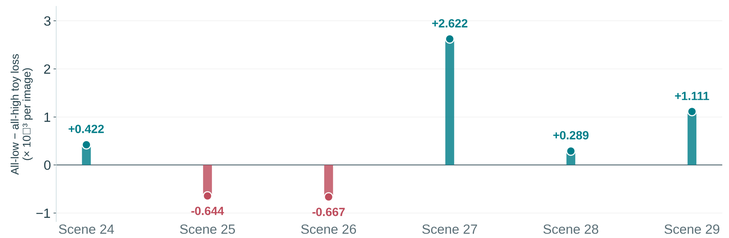
## 07 Nonadditivity and interaction

| 36.6% | r = 0.979 |
|---|---|
| lattice variation beyond additive stages | validation–test residual correlation |

An additive stage model leaves 36.6% of the across-mask test toy-loss variation unexplained; the prominent pairwise interaction involves stages 1 and 2. The two-anchor midpoint reproduces the exact four-stage Shapley ranking; actual placement also depends on hardware execution cost.

## 08 Six-group heterogeneity of precision benefit

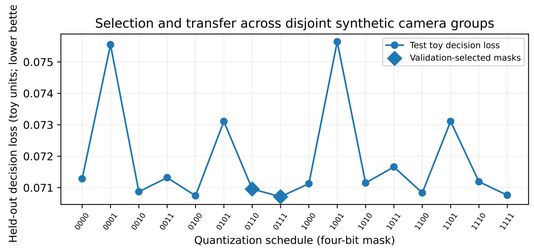
All-low minus all-high action loss (positive values favor higher precision). The six held-out groups differ in both direction and magnitude.




Four groups favor all-high; two favor all-low. The average effect should be read alongside the per-group variation.

## 09 Development selection and transfer

Schedules are selected using validation groups; the full lattice is then evaluated on held-out test groups.




**Reading the plot**

**Circles**

Test loss for each of the 16 masks.

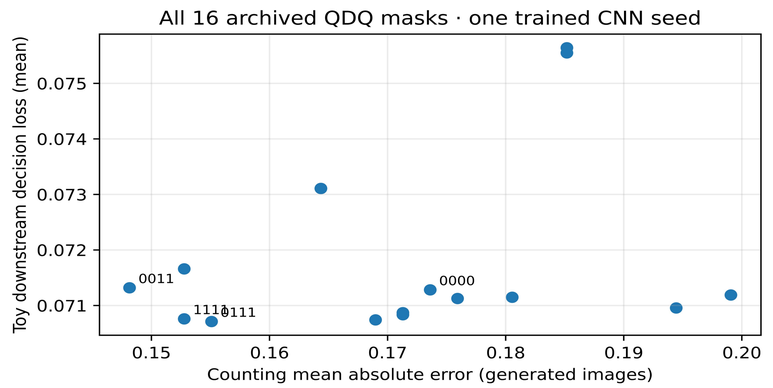
**Diamonds**

Validation-selected masks evaluated on the test groups.

Held-out toy decision loss (lower is better).

## 10 Precision schedules form a quality landscape

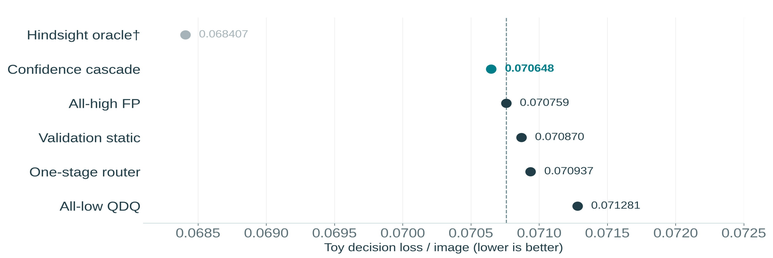
The 16-mask study reveals that count error and downstream loss rank precision placements differently (Spearman ρ = 0.180 across schedules).




The retrospective test-optimal count-MAE mask is 0011; the toy-decision-loss optimum is 0111. A posteriori best masks are descriptive and are not deployment-validated.

## 11 Comparison with policy baselines

Observed toy decision loss per image on the six held-out test groups; smaller is better. All policies use simulated precision, with overlapping uncertainty for leading deployable candidates.




†The hindsight oracle uses unavailable test information and is not deployable. The cascade reruns FP computation on 28.5% of images; its small improvement is not statistically established.

## 12 Energy accounting for adaptive inference

Compute savings depend on the whole inference path: the initial low-precision pass, the decision to escalate, optional high-precision reruns, and format conversions. Report joules per completed decision, not only the first model pass.

$$E_{\text{total}} = E_{\text{low pass}} + E_{\text{router}} + E_{\text{rerun}} + E_{\text{conversion}} + E_{\text{other}}$$

For building control, compare controller-compute and facility energy together, subject to comfort and reliability requirements.

## 13 Discussion and building-control evaluation

Across four CNN stages, precision interventions yield asymmetric decision changes, nonadditive interactions and scene-dependent effects. These experiments isolate a mechanism relevant to deciding when costly inference is worthwhile.

The building-control application calls for a closed-loop benchmark: occupancy or equipment sensing, adaptive inference, HVAC decisions, and joint measurement of controller energy, facility energy, thermal comfort, and reliability. The present evidence concerns visual-model precision rather than measured building energy savings.